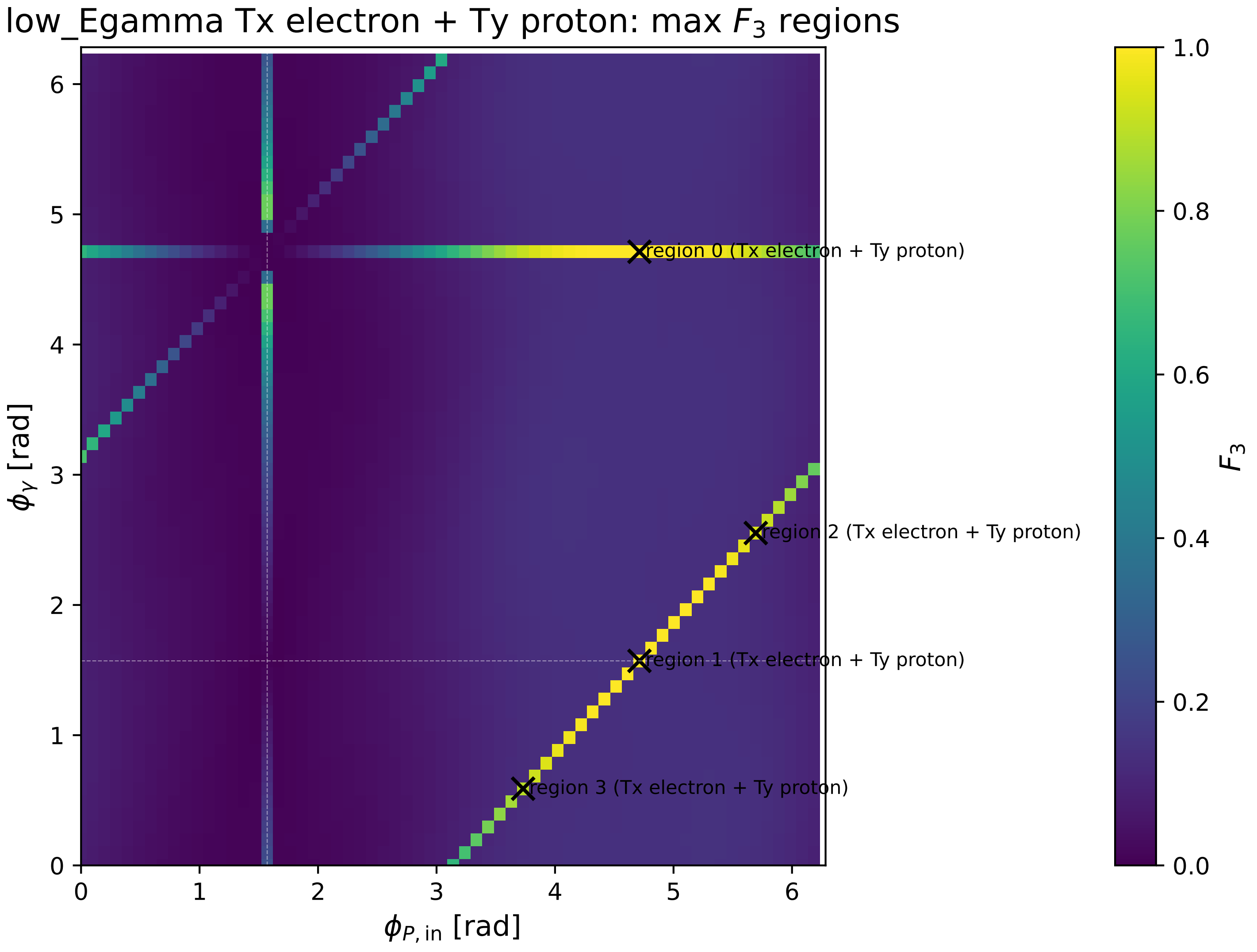
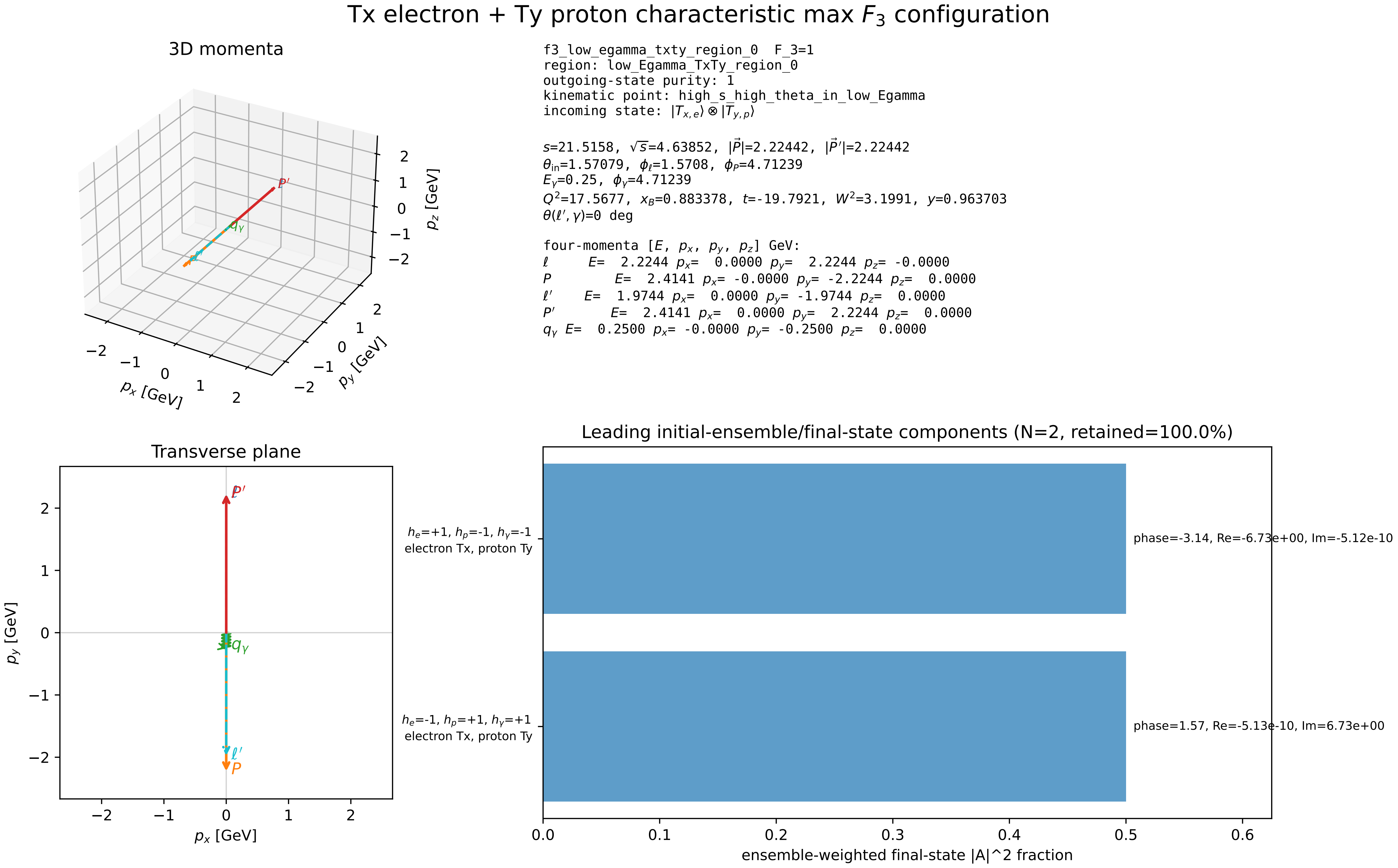
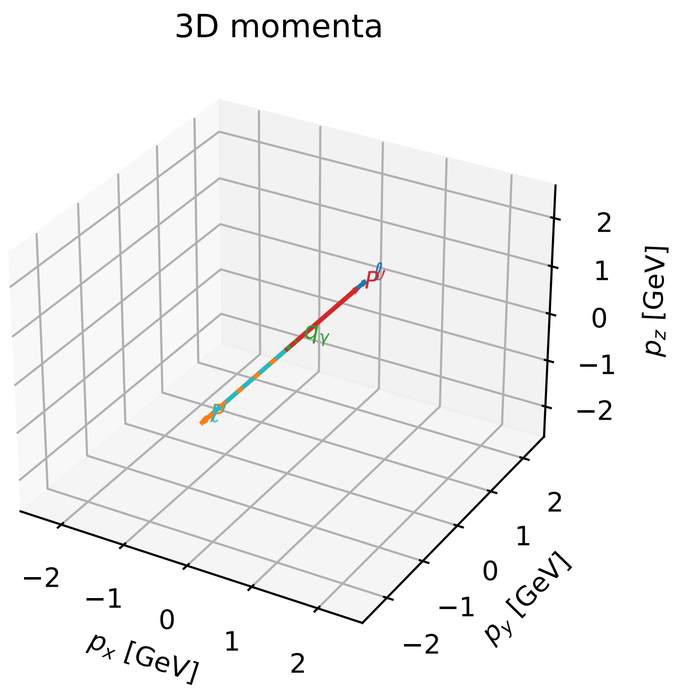


low_Egamma Tx electron + Ty proton: max F_3 regions
region 0 (Tx electron + Ty proton)
region 2 (Tx electron + Ty proton)
region 1 (Tx electron + Ty proton)
region 3 (Tx electron + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Tx electron + Ty proton characteristic max F_3 configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
f3_low_egamma_txty_region_0 F_3=1
region: low_Egamma_TxTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=4.71239
Q^2=17.5677, x_B=0.883378, t=-19.7921, W^2=3.1991, y=0.963703
$\theta(\ell',\gamma)$=0 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.9744 p_x= 0.0000 p_y= -1.9744 p_z= 0.0000
P' E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0000 p_y= -0.2500 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
P'
q_γ
ℓ'
P
Leading initial-ensemble/final-state components (N=2, retained=100.0%)
h_e=+1, h_p=-1, h_γ=-1
electron Tx, proton Ty
phase=-3.14, Re=-6.73e+00, Im=-5.12e-10
h_e=-1, h_p=+1, h_γ=+1
electron Tx, proton Ty
phase=1.57, Re=-5.13e-10, Im=6.73e+00
ensemble-weighted final-state |A|^2 fraction

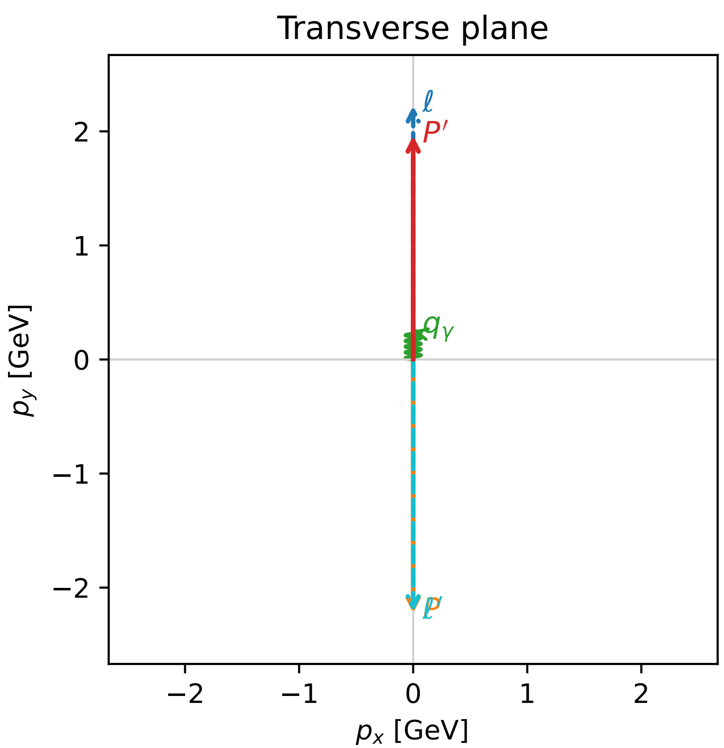
Tx electron + Ty proton characteristic max F_3 configuration

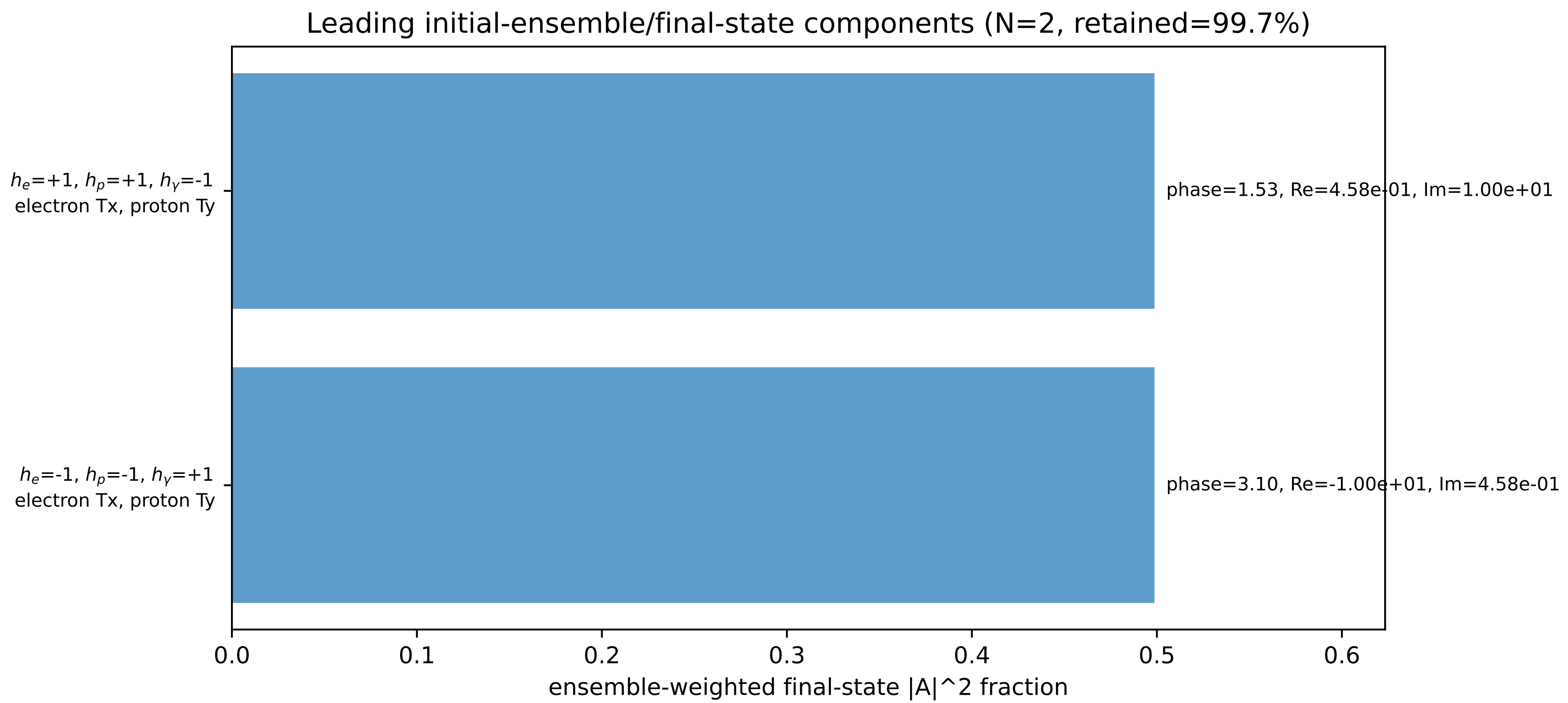


f3_low_egamma_txty_region_1 F_3=0.999993
region: low_Egamma_TxTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$

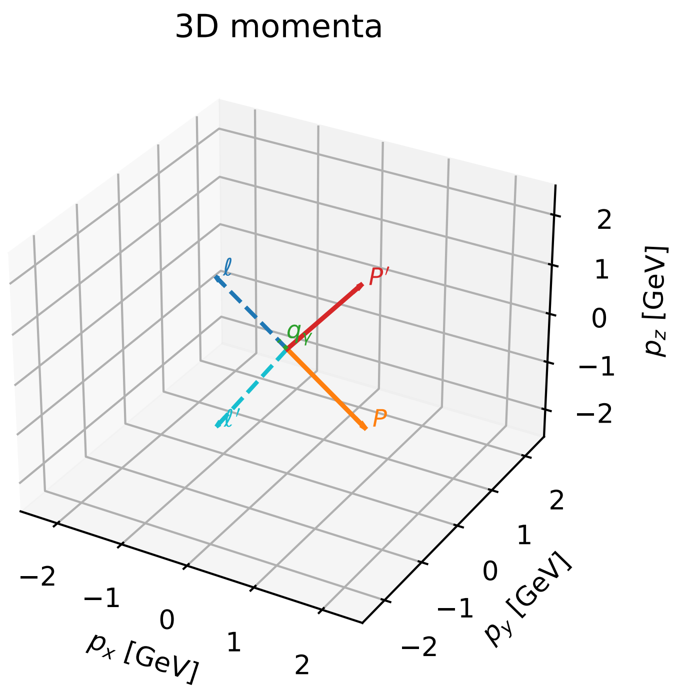
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.96296
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=1.5708
Q^2=19.6902, x_B=0.99463, t=-17.4772, W^2=0.986143, y=0.959318
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.2130 p_x= -0.0000 p_y= -2.2130 p_z= 0.0000
P' E= 2.1756 p_x= 0.0000 p_y= 1.9630 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0000 p_y= 0.2500 p_z= 0.0000





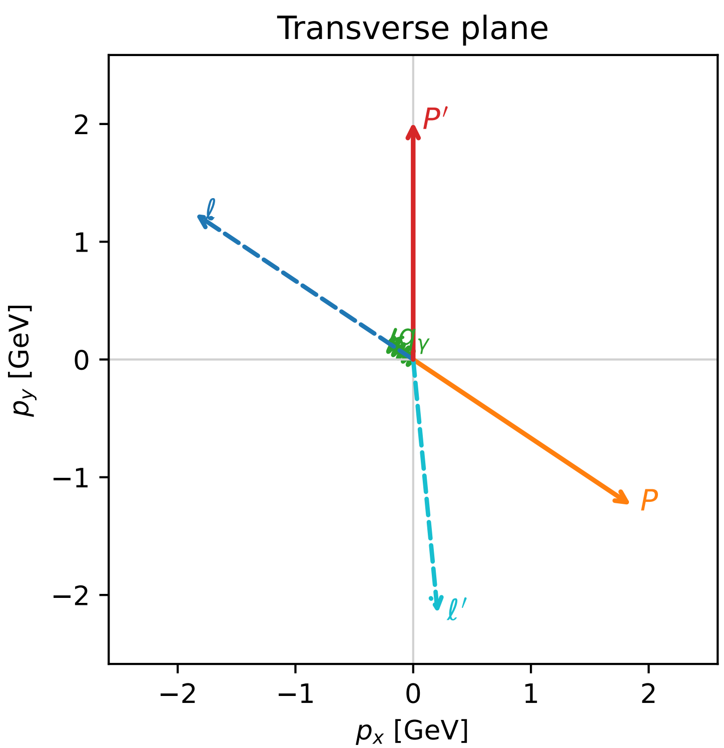
Tx electron + Ty proton characteristic max F_3 configuration

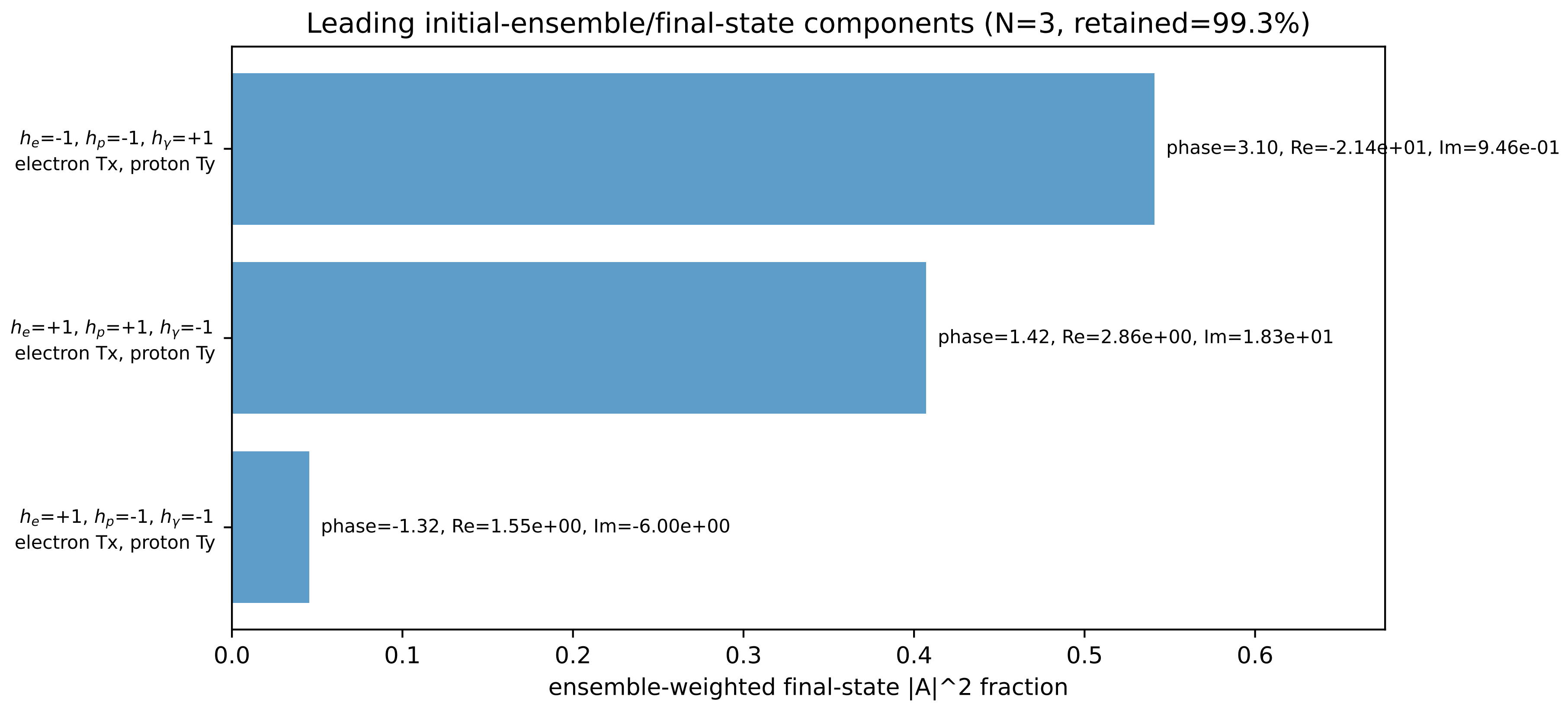


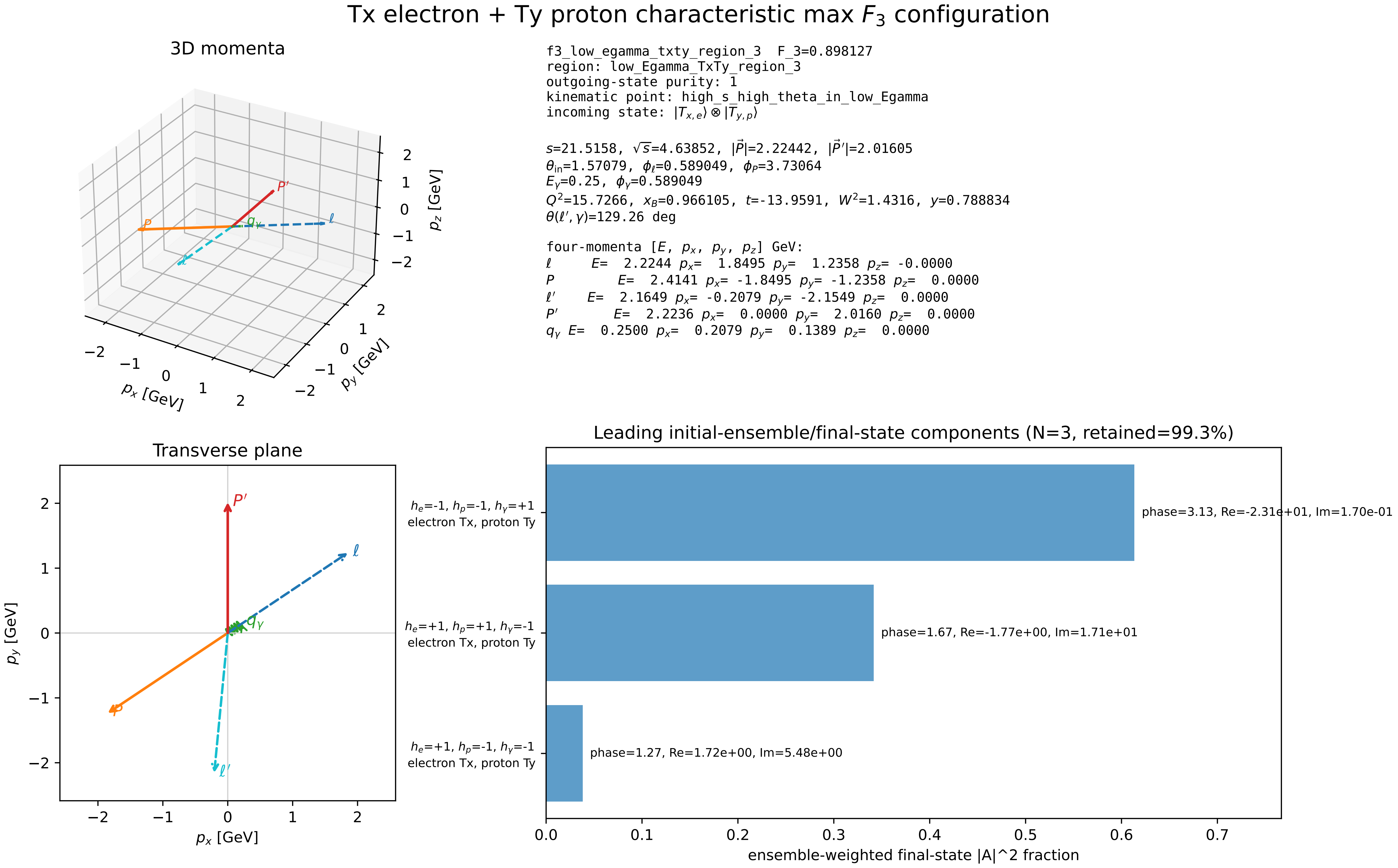
f3_low_egamma_txty_region_2 F_3=0.944209
region: low_Egamma_TxTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.01605
θ_{in}=1.57079, ϕ_ℓ=2.55254, ϕ_P=5.69414
E_γ=0.25, ϕ_γ=2.55254
Q^2=15.7266, x_B=0.966105, t=-13.9591, W^2=1.4316, y=0.788834
$\theta(\ell', \gamma)$=129.26 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= 1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.1649 p_x= 0.2079 p_y= -2.1549 p_z= 0.0000
P' E= 2.2236 p_x= 0.0000 p_y= 2.0160 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2079 p_y= 0.1389 p_z= 0.0000






Tx electron + Ty proton characteristic max F_3 configuration
3D momenta
f3_low_egamma_txty_region_3 F_3=0.898127
region: low_Egamma_TxTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.01605
θ_{in}=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=0.25, ϕ_γ=0.589049
Q^2=15.7266, x_B=0.966105, t=-13.9591, W^2=1.4316, y=0.788834
$\theta(\ell', \gamma)$=129.26 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= -1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.1649 p_x= -0.2079 p_y= -2.1549 p_z= 0.0000
P' E= 2.2236 p_x= 0.0000 p_y= 2.0160 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2079 p_y= 0.1389 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading initial-ensemble/final-state components (N=3, retained=99.3%)
h_e=-1, h_p=-1, h_γ=+1
electron Tx, proton Ty
phase=3.13, Re=-2.31e+01, Im=1.70e-01
h_e=+1, h_p=+1, h_γ=-1
electron Tx, proton Ty
phase=1.67, Re=-1.77e+00, Im=1.71e+01
h_e=+1, h_p=-1, h_γ=-1
electron Tx, proton Ty
phase=1.27, Re=1.72e+00, Im=5.48e+00
ensemble-weighted final-state |A|^2 fraction